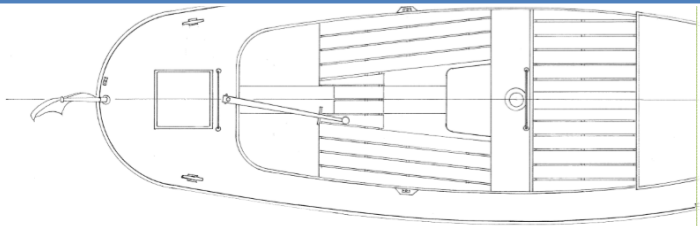


Milford 20

DESIGN OVERVIEW



Neville Watkinson

Milford Boats

Sharpie schooner yacht

Milford 20

People seem drawn to both the simplicity of sharpies and the streamlined aesthetic of the shape which looks sleek and stylish yet is very traditional.

The *Milford 20* is intended as a modern representation of the New Haven boats from the shores of Long Island Sound, Connecticut.

Originally built to serve the Oyster fishing industry, Sharpies soon migrated down the eastern seaboard, inland to the Great lakes, across the Atlantic to France and into recreational pursuits.



DESIGN BRIEF

The Milford 20 has been designed to satisfy a number of criteria:

- A narrow, easily driven hull that will perform well with modest sail area
- Trailer-able, and easily launched / retrieved.
- Relatively light so it may be towed by a small car
- A vessel built primarily for a couple, or family with young children.
- Plywood construction to suit amateur builders.
- Enough detail to lend an air of elegance in keeping with the *Floridays* design which inspired it.
- A simple rig with un-stayed masts and schooner sail plan.

Design commentary:

There are no shortage of sharpie designs on the market, so why this one? The answer lies partly in the aesthetic appeal of the Milford 20. The elements that set it apart are the counter stern, the cabin with its elliptical port-lights, the trim detail that would typically be found on a larger yacht and the schooner rig. It's a classic design where the styling is more important than maximising the use of available space. These features do make this yacht more time consuming to build than say a transom stern counterpart, but the time spent will reward the owner with a lovely vessel that will be the envy of many.

Construction follows traditional plank on frame methodology, but using plywood for the hull sheathing. The process is well within the capabilities of an amateur with basic woodworking skills and a modest collection of tools. Finishing the cockpit coamings and cabin detail with rounded corners represents a bit more challenge, and will require access to a greater range of tools, but the process is documented in the build manual. Of course the builder has the option of simplifying these details with a commensurate saving in time and cost.

The Milford 20 is quite a small boat for its length. Sharpies have a fairly shallow hull and the crew sit on the cockpit floor boards. Because of this low seating position the boat feels very secure, and is quite dry underway. It's not like sailing a dinghy. Being such an easily driven hull light air sailing performance is especially pleasing. The boat will make way and respond to the helm under conditions of near calm.

Sharpies need to be sailed flat, and as the wind picks up the sails are reefed early. The Milford 20 fair flies along when it's blowing between 15 and 20 knots, with both sails double reefed. Under these conditions she typically heels about 10 – 15 degrees while maintaining 6 knots boat speed.

A frequently asked question relates to camp cruising. There is sleeping length available under the cabin and in the cockpit, however for such a shallow draft vessel it's easy to park on a beach, and carrying a tent and camping equipment is possibly a more practical solution.

Plans:

The plans are only available to metric dimensions. Most have been drawn to imperial A1 size (820 x 597mm) and are at a scale of 10:1.

If building from the pdf format, it would be helpful to get the construction and section plans printed out, as these are referred to most often. The balance can probably be viewed on a computer screen as needed. The following plans are included:

1. General arrangement
2. Schooner sail plan
3. Ketch sail plan
- 3 Construction plan & offsets
4. Mid ships section plan
5. Stern frame detail
6. Centreboard detail with full size plans for ballast
7. Spars and fittings
8. Bill of Materials – hull construction & plywood sheet layout
9. Bill of Materials – Rig & fittings
10. Cabin trunk layout
11. Inboard engine & propeller shaft detail
12. Trailer plan & scantlings



Sails:

Sails are available through the North Sail loft network worldwide, or can be made by your local sail maker from the plans supplied.

Engine:

The prototype was built with a 6hp. Air cooled Honda industrial engine (model GX-200) mounted inboard under the bridge deck. This works very well in practice. Plans have recently been drawn covering this, and would easily be adaptable to other types of motive power.

Construction:

The Milford 20 follows fairly conventional construction practice. The hull is built upside down over a building frame. There are no temporary moulds, the hull is built around the permanent frames, bulkheads and longitudinal members. There are three bulkheads, one near the bow, and two water-tight bulkheads. One positioned amidships and one just ahead of the rudder post. This aft bulkhead encloses the front of the lazarette, forming a potentially watertight compartment (provided the hatch is sealed). This volume contributes 180 kg (400 lbs.) of buoyancy. The addition of some foam throughout the hull would render enough positive buoyancy to support the hull and its occupants in the event of a capsize. Hull weight when turning over to complete the deck is around 100 kg. (220 lbs.).

The centreboard is placed between the masts. The midships bulkhead forms the front face of a bridge deck, which incorporates the main mast partners. This has the effect of creating two cockpits. A working cockpit behind the bridge deck from where the boat will be sailed, and a more intimate space ahead which extends under the coach house, both cockpits are long enough to sleep in.

The counter stern is cold moulded from 3 mm. ply, or it could be strip planked. The rounded cockpit coamings and the cabin front are similarly laminated.

The centreboard is laminated with timber and lead to provide 75 kg. of ballast (165 lbs.). This weight below the waterline is an aid to stability and helps dampen the boats motion in a seaway.

SPECIFICATIONS:

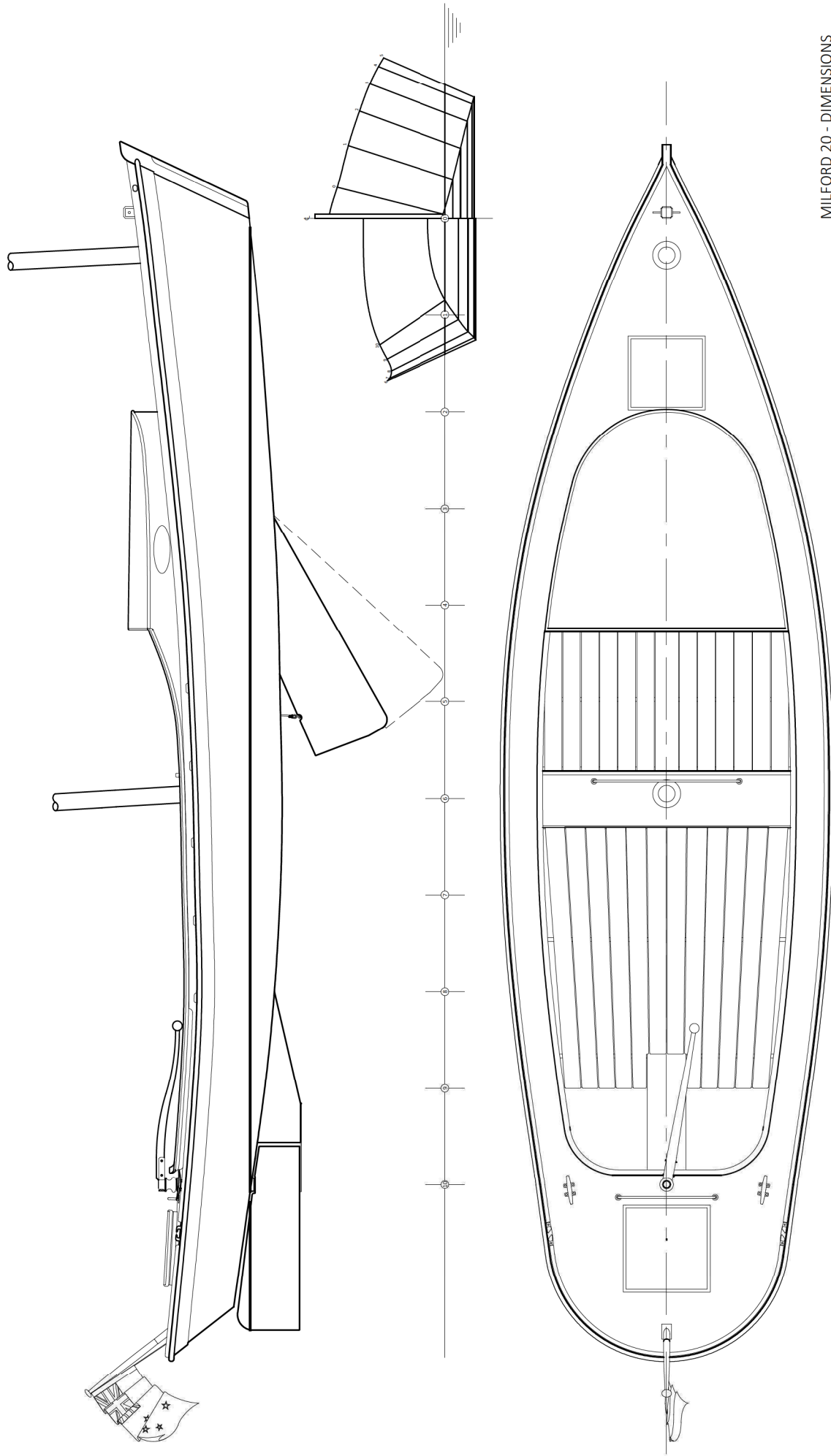
Key dimensions:

Please note that the plans are only available in metric form

Length overall	6,250 mm	20' 6"
Length on waterline	5,000 mm	16' 5"
Beam overall	1,720 mm	5' 8"
Beam to outside of plank	1,680 mm	5' 6"
Waterline beam	1,406 mm	4' 7"
Draft – board up	245 mm	10"
Draft – board down	1,000 mm	3' 3"
Hull depth	160 mm	6"
Forward overhang	360 mm	1' 2"
Aft overhang	890 mm	2' 11"
Freeboard - bow	600 mm	2' 0"
- Least	280 mm	11"
- Stern	425 mm	1' 5"
Sail area	44 sq. M.	145 sq. ft.
Displacement	618 kg	1360 lbs
Hull weight (including centre board)	315 kg	695 lbs
Weight for 1" immersion	97 kg	215 lbs
Beam to length ratio	3.63	3.63
Waterline beam to load waterline ratio	3.56	3.56
Displacement to length ratio	111	111
Prismatic coefficient	0.57	0.57
Towing weight (typical – depends on trailer)	600 kg	1320 lbs

Scantlings:

Keelson	200 x 20 mm	8 x ¾"	Timber
Chine logs	65 x 20 mm	2 ½ x ¾"	Timber
Sheer clamps	45 x 20 mm	2 x ¾"	Timber
Frames	45 x 20 mm	2 x ¾"	Timber
Floors	65 x 20 mm	2 ½ x ¾"	Timber
Bulkheads (3 of)	12 mm	½"	Marine ply
Bottom planking	12 mm	½"	Marine ply
Side planking	9 mm	¾"	Marine ply
Deck planking	6 mm	¼"	Marine ply



MILFORD 20 - DIMENSIONS	
LENGTH OVERALL:	6,250 mm
WATERLINE LENGTH:	5,000 mm
BEAM:	1,720 mm
DRAFT:	245 / 1000 mm
DISPLACEMENT:	620 kg
SAIL AREA:	14 sq.m. 150 sq.ft.



